An exponential-hierarchy oracle gives infinitely-often linear circuits for exponential Merlin-Arthur computation

Shuichi Hirahara, Zhenjian Lu, Hanlin Ren

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Editorial note. Full author official mathematical body retained, including the main statement18 and proof given as detailedLemma62. Proof establishes the stronger promise-class statement; no supporting lemma is separately counted. Published set inclusions and probability thresholds are transcribed without mathematical corrections.

Allowed standard prerequisites (editorial).

- Approximate counting in the polynomial hierarchy and standard search-to-decision reductions. Original author Theorems23/24 (with author proof sketch24) retained; these are standard counting/complexity inputs, not counted results

Definitions and notations (original Sections 1.1, 2.1). For a complexity class $\mathfrak{C}$, a complexity-theoretic statement is $\mathfrak{C}$-relativizing if it is true relative to every oracle $\mathcal{O} \in \mathfrak{C}$.

Definition 20. A language L is in EXPH if there is a constant k, an exponential bound $e(n) = 2^{\mathrm{poly}(n)}$, and a $\mathrm{poly}(e(n))$-time algorithm M such that, for every input $x \in \{0,1\}^*$,

$$x \in L \iff \exists y_1 \forall y_2 \exists y_3 \forall y_4 \dots \exists y_k,\ M(x, y_1, \dots, y_k) = 1,$$

where $y_i \in \{0,1\}^{e(n)}$ for each $i \in [k]$. Equivalently, a language is in EXPH if it can be decided in exponential $(2^{\mathrm{poly}(n)})$ time with access to a PH oracle.

Pseudodeterministic algorithms. A pseudodeterministic algorithm is a probabilistic algorithm that returns a fixed output with high probability on any given input. We will also consider pseudodeterministic oracle (or advice-taking) algorithms. Such an algorithm will maintain its pseudodeterministic behavior (i.e., outputting a fixed answer with high probability for any input) when a correct oracle (or advice) is given.

Dense property (original Section 1.3.2). A property $Q \subseteq \{0,1\}^*$ is dense if for every input length $n \in \mathbb{N}$, we have that $|Q \cap \{0,1\}^n| \geq 2^n / \mathrm{poly}(n)$.

Theorem 23 [68]. For every oracle $\mathcal{O}$, $\mathsf{BPP}^{\mathcal{O}} \subseteq (\Sigma_2^p)^{\mathcal{O}}$.

6.2 Bounded-Relativization Barriers for Circuit Lower Bounds

We present an EXPH-relativization barrier for proving an almost-everywhere version of Santhanam's lower bound. The oracle construction is based on [13]. Although the construction in [13] does not seem to be in EXPH, by modifying the construction using approximation counting in PH, we obtain an EXPH-computable oracle under which $\mathsf{MA}/1$ is computable by linear-sized circuits infinitely often.

Theorem24 (Approximate Counting in Quasi-Linear-Time Hierarchy). There is a $\Sigma_5\mathsf{TIME}[n \cdot \mathrm{polylog}(n)]$ machine M such that on input (φ, K), accepts if φ has at least K satisfying assignments, and rejects if φ has at most $K/2$ satisfying assignments.

Proof Sketch. Using universal hashing [69], we can show that there is a $\mathsf{BPTIME}[n \cdot \mathrm{polylog}(n)]^{\mathsf{SAT}}$ machine M' that approximates the number of satisfying assignments of φ'. Let H be a family of pairwise-independent hash functions whose output range is $\{0,1\}^{\lceil \log K \rceil}$, that is computable in near-linear time. (See, e.g., [72, Problem3.3,(2)] for an example based on Toeplitz matrices.) The machine M' randomly samples $h \sim H$ and $r \sim \{0,1\}^{\lceil \log K \rceil}$, and uses the SAT oracle to decide whether there is a satisfying assignment x of φ' such that $h(x) = r$. Finally, the lemma follows from the fact that for every oracle $\mathcal{O}$, $\mathsf{BPTIME}^{\mathcal{O}}[t] \subseteq \Sigma_3\mathsf{TIME}[t \cdot \mathrm{polylog}(t)]^{\mathcal{O}}$ [74]. □

Theorem18 (introduction). There is an oracle $\mathcal{O} \in \mathsf{EXPH}$ such that pr-$\mathsf{MATIME}^{\mathcal{O}}[2^n] \subseteq$ i.o.$\mathsf{SIZE}^{\mathcal{O}}[O(n)]$.

Lemma62. There exists an oracle $\mathcal{O} \in \mathsf{EXPH}$ such that

$$\mathsf{MATIME}^{\mathcal{O}}[2^n]/1 \subseteq \text{i.o.}\,\mathsf{SIZE}^{\mathcal{O}}[O(n)].$$

Proof. We prove pr-$\mathsf{MATIME}^{\mathcal{O}}[2^n] \subseteq$ i.o.$\mathsf{SIZE}^{\mathcal{O}}[O(n)]$ for some oracle $\mathcal{O}$. We enumerate all the 2^n-time randomized machine $M_1, M_2, \ldots$, where each M_i takes an input x of length n and a certificate $y \in \{0,1\}^{2^n}$ and runs in time 2^n. Note that any MA-type algorithm that runs in time 2^n can be simulated by M_i for some i. Let c be some universal constant ($c := 5$ suffices). Let $R_n := \{0,1\}^{cn}$. The input to $\mathcal{O}$ is of the form (r, i, x), where $r \in R_n$, $i \in [n]$, and $x \in \{0,1\}^n$ for some $n \in \mathbb{N}$. For any $r \in \{0,1\}^{cn}$, let S_r be the set of (r, i, x) such that $i \in [n]$ and $x \in \{0,1\}^n$. We will construct an oracle $\mathcal{O}$ and a family of "advice strings" $r_n^* \in R_n$ for infinitely many n such that

1. if $\Pr_{M_i}[M_i^{\mathcal{O}}(x,y) = 1] \geq \frac{3}{4}$ for some y, then $(r_n^*, i, x) \in \mathcal{O}$, and
2. if $\Pr_{M_i}[M_i^{\mathcal{O}}(x,y) = 1] \leq \frac{1}{4}$ for any y, then $(r_n^*, i, x) \notin \mathcal{O}$.

Assuming this, it is easy to construct an $\mathcal{O}$-oracle linear-size circuit that simulates M_i as follows. The circuit takes r_n^* as hard-wired input and accepts an input $x \in \{0,1\}^n$ if and only if $(r_n, i, x) \in \mathcal{O}$. Here is the construction of $\mathcal{O}$. We start with $\mathcal{O} := \emptyset$. Some pairs (i, x) will be marked "forced", meaning that M_i

accepts on input x. Some strings r will be marked "frozen", meaning that the behavior of $\mathcal{O}$ on inputs in S_r will not be changed after r is frozen. Initially, no pair is forced and no advice string is frozen. We start with Stage $n := 1$. In Stage n, we construct an oracle as follows.

Stage n. Consider the following condition, which we call $(*)_\theta$ for a threshold $\theta \in (0,1)$.

> There exist an unfrozen string $r \in R_n$, an unforced pair $(i,x) \in [n] \times \{0,1\}^n$, an oracle $B \subseteq S_r$, and a certificate $y \in \{0,1\}^{2^n}$ such that
> $$\Pr_{M_i}[M_i^{\mathcal{O} \cup B}(x,y) = 1] \geq \theta,$$
> where the probability is taken over the coin flip of the randomized machine M_i.

We need to argue that the final oracle $\mathcal{O}$ can be computed in EXPH. It may not be possible to check $(*)_{3/4}$ exactly in EXPH, but using approximate counting in PH (Theorem23), we can check whether a promise variant of $(*)_{3/4}$ is satisfied or not in EXPH. Specifically, by Theorem23, there exists an algorithm in PH that, given as input n and $\mathcal{O}$ (which can be encoded as a binary string of length $2^{O(n)}$), accepts if $(*)_{3/4}$ holds, and rejects if $(*)_{1/2}$ does not hold. By the standard search-to-decision reduction, we obtain a PH-oracle polynomial-time algorithm S that outputs $r \in R_n$, $(i,x) \in [n] \times \{0,1\}^n$, $B \subseteq S_r$, and $y \in \{0,1\}^{2^n}$ that satisfy $(*)_{1/2}$ if $(*)_{3/4}$ holds, and outputs $\perp$ if $(*)_{1/2}$ does not hold. (Note that S runs in time $2^{O(n)}$ on inputs of length $2^{O(n)}$ with a PH oracle.) While the search algorithm S outputs a certificate (r,i,x,B) for $(*)_{1/2}$ on input $(n,\mathcal{O})$ (instead of $\perp$), we do the following. Update $\mathcal{O} := \mathcal{O} \cup B$. Let r be frozen and let (i,x) be forced. Let R_θ be the set of $r' \in R_n$ such that (r',i',x') is queried during the computation of $M_i^{\mathcal{O}}(x,y)$ for some (i',x') with probability at least θ over the internal randomness of M_i. By Theorem24, some set A such that $R_{2^{-2^n}} \subseteq A \subseteq \{0,1\}^* \setminus R_{2^{-2^{n-1}}}$ can be enumerated in EXPH. Note that $|A| \leq 2^{2n+1} \cdot 2^n$ because M_i can make at most 2^n queries on each computation path. Let r' be frozen for every $r' \in A$. If S outputs $\perp$ (and thus $(*)_{3/4}$ does not hold), then let $r_n^* \in R_n$ be the first string that is not frozen. (We will later claim that such a string r_n^* exists.) Then, we add to $\mathcal{O}$ all the tuples (r_n^*,i,x) such that $(i,x) \in [n] \times \{0,1\}^n$ is forced. This completes the description of Stage n. Then we move on to the next stage $n' := 2^n + 1$, so that the construction in each stage is independent. It is evident from the construction that $\mathcal{O} \in \mathsf{EXPH}$. Fix any n and consider Stage n. We claim that there exists a string $r_n^* \in R_n$ that is not frozen at the end of Stage n. We say that $(*)$ is satisfied if S does not output $\perp$. Observe that the number of times that $(*)$ is satisfied is at most $n2^n$. The reason is that there are at most $n2^n$ pairs $(i,x) \in [n] \times \{0,1\}^n$, and each time $(*)$ is satisfied, (i,x) becomes forced. Each time $(*)$ is satisfied, at most $1 + 2^{2n+1} \cdot 2^n \leq 2^{3n+2}$ strings $r \in R_n$ can be frozen. Thus, the number of frozen strings in R_n in Stage n is at most $n2^n \cdot 2^{3n+2}$. Since there are at most n stages before Stage n, in total, there are at most $n \cdot n2^n \cdot 2^{3n+2} < 2^{5n} \leq |R_n|$ frozen strings in R_n. (Here, we used that $c := 5$.) Thus, there exists some unfrozen string $r_n^* \in R_n$, which shows that the construction of $\mathcal{O}$ is well defined. It remains to show that for every pair $(i,x) \in [n] \times \{0,1\}^n$,

1. if $\Pr_{M_i}[M_i^{\mathcal{O}}(x,y)=1] \geq \frac{3}{4}$ for some y, then (i,x) is forced (and thus $(r_n,i,x) \in \mathcal{O}$), and
2. if $\Pr_{M_i}[M_i^{\mathcal{O}}(x,y)=1] \leq \frac{1}{4}$ for any y, then (i,x) is not forced (and thus $(r_n,i,x) \notin \mathcal{O}$).

Fix any n and consider any unforced pair $(i,x) \in [n] \times \{0,1\}^n$. We claim that for every $y \in \{0,1\}^{2^n}$, it holds that

$$\Pr_{M_i}[M_i^{\mathcal{O}}(x,y)=1] < \tfrac{3}{4}.$$

Fix any $y \in \{0,1\}^{2^n}$. Let $\mathcal{O}_n$ be the oracle $\mathcal{O}$ right after the while loop of Stage n. Since $(*)_{3/4}$ is not satisfied at that point, for every $r \in R_n$ and for every oracle $B \subseteq S_r$,

$$\Pr_{M_i}[M_i^{\mathcal{O}_n \cup B}(x,y)=1] < \tfrac{3}{4}.$$

Let $B \subseteq S_{r_n^*}$ be the set of strings added to $\mathcal{O}$ at the end of Stage n. Then, the final oracle $\mathcal{O}$ coincides with $\mathcal{O}_n \cup B$ on any inputs of length at most 2^n; thus, we obtain

$$\Pr_{M_i}[M_i^{\mathcal{O}}(x,y)=1] = \Pr_{M_i}[M_i^{\mathcal{O}_n \cup B}(x,y)=1] < \tfrac{3}{4}.$$

Next, fix any n and consider any forced pair $(i,x) \in [n] \times \{0,1\}^n$. Let $\mathcal{O}_{i,x}$ be the oracle right after (i,x) is forced and y be the certificate that satisfies $(*)$. Then, we have

$$\Pr[M_i^{\mathcal{O}_{i,x}}(x,y)=1] \geq \tfrac{1}{2}.$$

We claim that this probability does not decrease by $\frac{1}{4}$ even if we replace $\mathcal{O}_{i,x}$ with the final oracle $\mathcal{O}$. After (i,x) is forced, $(*)$ can be satisfied at most $n2^n$ times, and each time $(*)$ is satisfied, at most one S_r whose subset is added to $\mathcal{O}$. Thus, there are at most $n2^n$ unfrozen strings $r \in R_n$ such that some strings in S_r are added to $\mathcal{O}$. Note that any string r that is queried by M_i with probability 2^{-2n} is frozen right after (i,x) is forced. Thus, each unfrozen string can decrease the probability that $M_i^{\mathcal{O}}(x,y)$ accepts by 2^{-2n}. In total, the unfrozen strings can affect the probability of acceptance by $n2^n \cdot 2^{-2n} < \frac{1}{4}$. Thus, we obtain

$$\Pr[M_i^{\mathcal{O}}(x,y)=1] > \tfrac{1}{2} - \tfrac{1}{4} \geq \tfrac{1}{4}.$$

□

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